innovate achieve lead

BITS Pilani
Pilani Campus

CE F230
Civil Engineering Materials

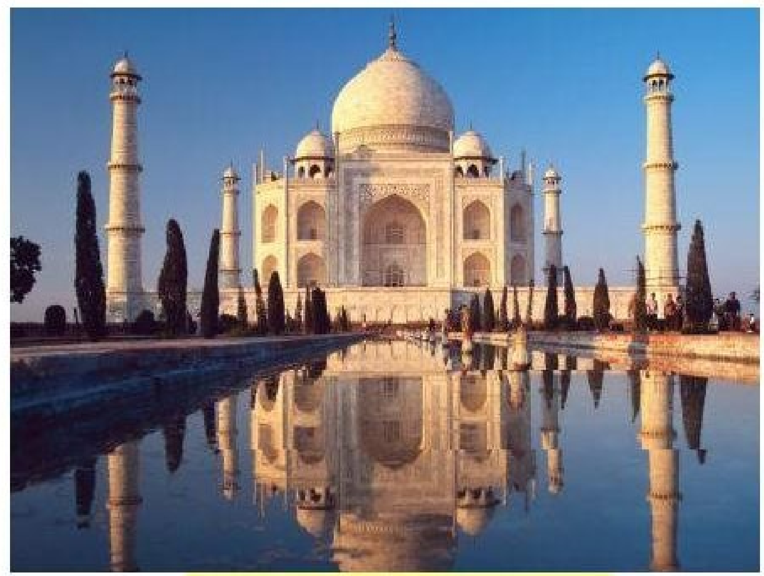
Aggregates

Stone As A Building Material

stone pylons for an old bridge

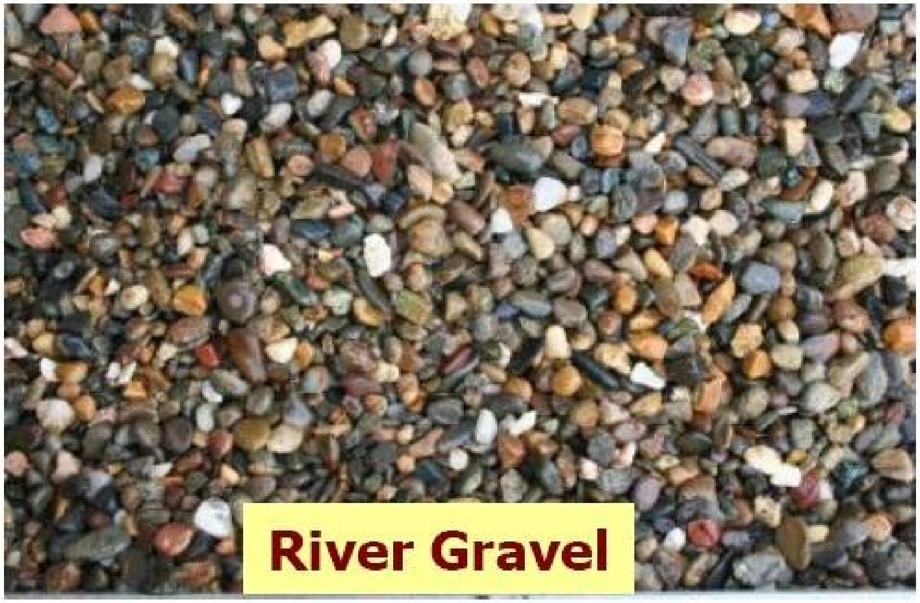
Church building

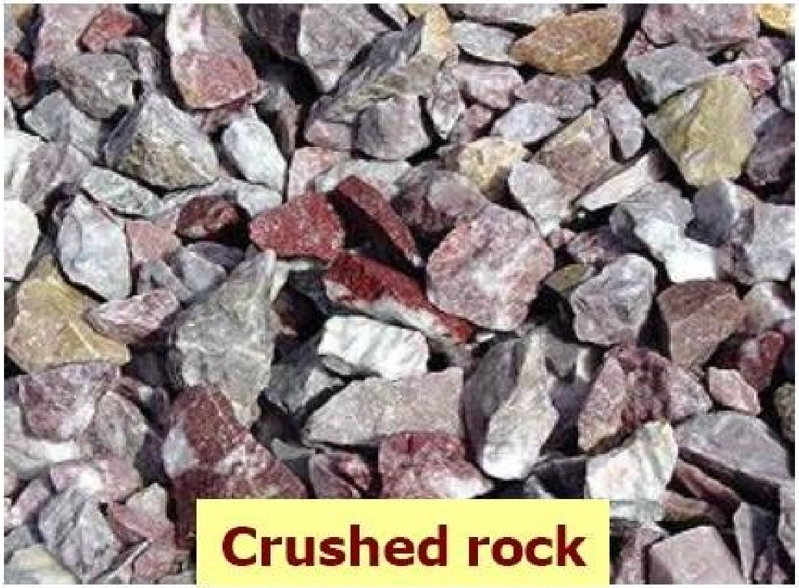
Machu Picchu – Incas structure

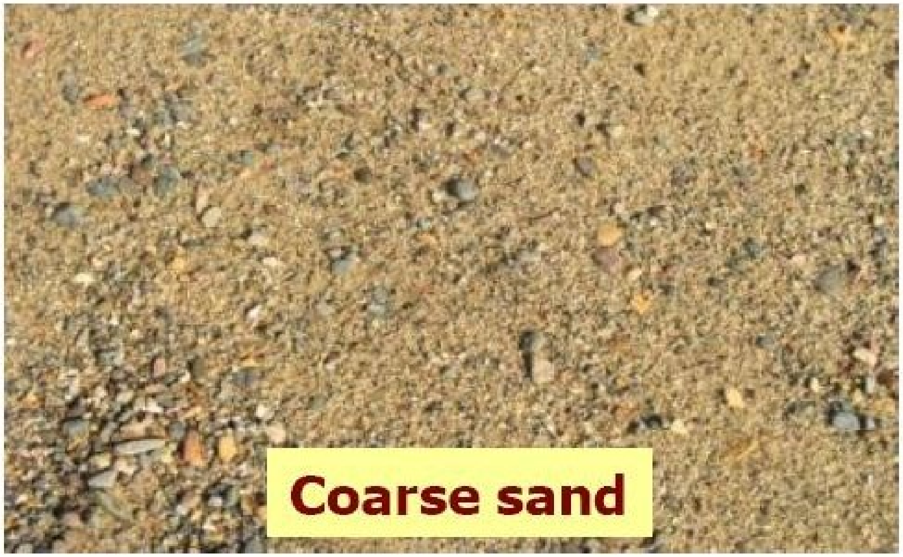


Taj Mahal - India

Aggregates for Concrete Mixes







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Roles of Aggregates in Concrete

- ECONOMICAL FILLER
  - (60 - 80% BY VOLUME)
- PROVIDES DIMENSIONAL STABILITY
  - AGGREGATE – RIGID AND STRONG
  - CEMENT PASTE – RELATIVELY SOFT; SHRINKING, SWELL AND CREEP

Classification of Aggregate: Size

- COARSE AGGREGATE
  - RIVER GRAVEL; BASALT, GRANITE, LIMESTONE, BLAST-FURANCE SLAG
- FINE AGGREGATE
  - RIVER SAND, QUARRY SAND, DUNE SAND, PIT SAND

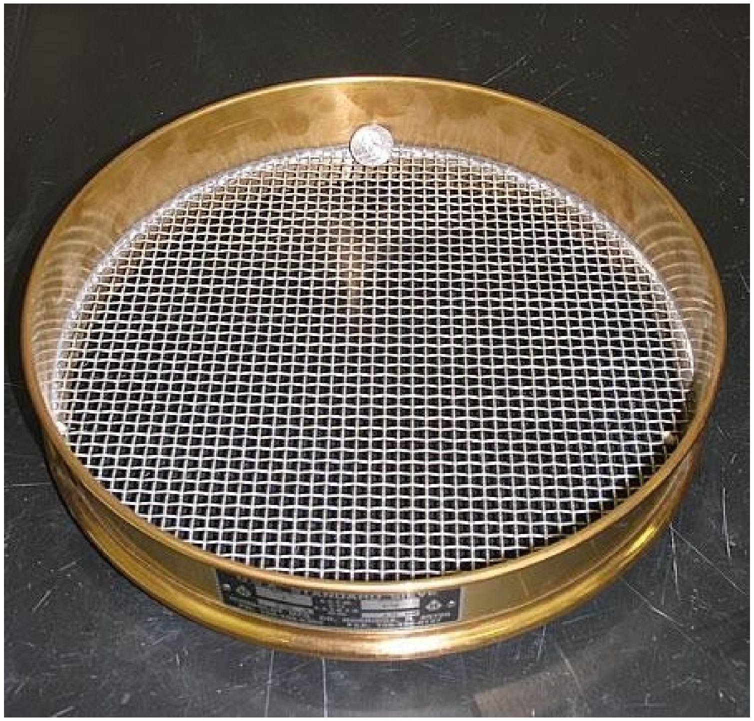
River gravel aggregate

Natural sand

Granite aggregate

Aggregate Sizes

- **Coarse** aggregate material retained on a sieve with 4.75 mm openings
- **Fine** aggregate material passing a sieve with 4.75 mm openings



- Maximum aggregate size – the largest sieve size that allows all the aggregates to pass
- Nominal maximum aggregate size – the first sieve to retain some aggregate, generally less than 10%

Classification of Aggregate: Source

- **Natural**
  - Natural sand & gravel pits, river rock
  - Quarries (crushed)
  - Geological classification
    - Igneous: Basalt; Granite
    - Sedimentary: Limestone; Sandstone
    - Metamorphic: Marble; Quartzite

River gravel

Crushed rock

basalt

Limestone

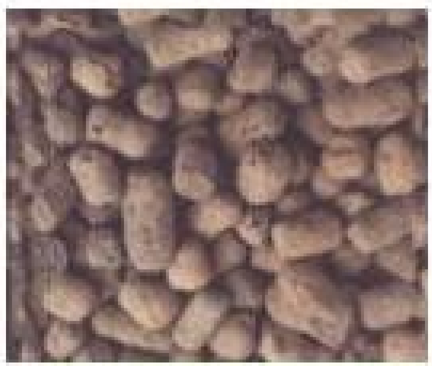
granite

sandstone

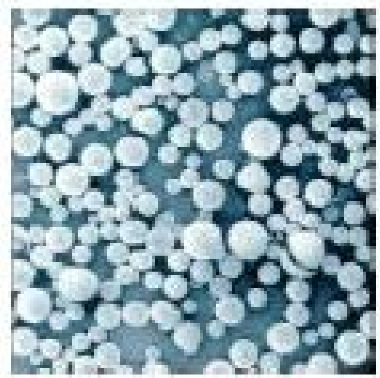
Classification of Aggregate: Source

- **Manufactured & recycled materials:**
  - Pulverized concrete & asphalt
  - Steel mill slag
  - Expanded shale
  - Styrofoam

Recycled concrete aggregate



Expanded shale



Expanded polystyrene

Porous aggregate

Aggregate Properties

Typical source properties relevant for PCC mix design

- Shape and texture
- Soundness and durability
- Hardness and abrasion resistance
- Absorption
- Specific gravity
- Strength
- Gradation
- Cleanness and deleterious materials
- Alkali-aggregate reactivity

Sampling Aggregates

- **Random** and **representative** of entire stockpile
  - Sample from entire width of conveyor belts at several locations
  - Sample from top, middle, and bottom of stockpile at several locations around stockpile diameter
  - Use larger sample for testing larger max. size
- Sample splitting or quartering
  - To reduce sample size from large stockpile to small 1-5 kg sample

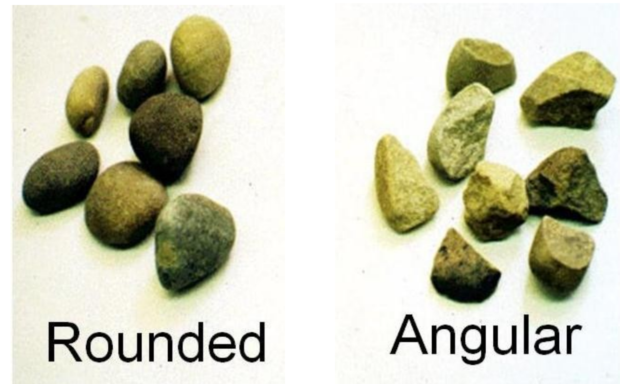
Random sampling locations

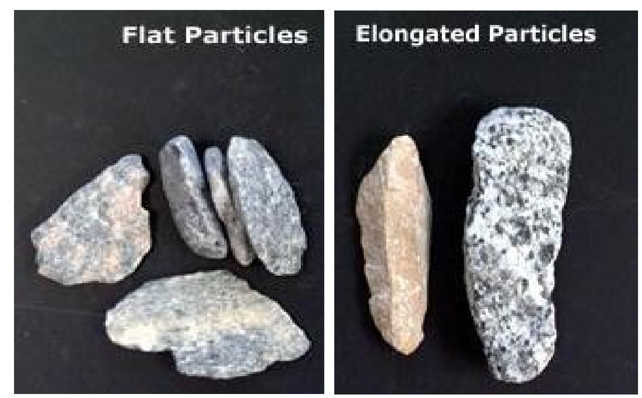
Sample Splitter

Quartering

Particle Shapes of Aggregates

- **Rounded**: river gravel
  - Better packing, lowest voids ratio 33%
  - Less interlocking between particles
- **Angular**: crushed rock
  - Loose packing, higher voids ratio
  - Interlocking between particles is good
- **Flaky**: small thickness
- **Elongated**: length considerable
- **Flaky & elongated**: thin & long
  - Bad for concrete durability because of easy breakage and difficulty compacting
  - Should be restricted to 10-15% in concrete design

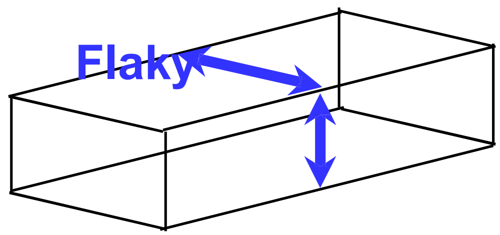




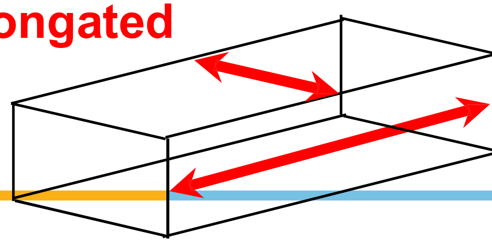
Particle Shape Characterization

- **Angularity number**
  - = voids ratio – 33
  - Higher the number, more angular the aggregate
  - The range for practical aggregate is between 0 (rounded) and 11 (angular)

- Particle is **flaky** if its smallest dimension (thickness) is less than 60% of the “middle” dimension



- Particle is **elongated** if its largest dimension (length) is more than 1.8 times of the “middle” dimension

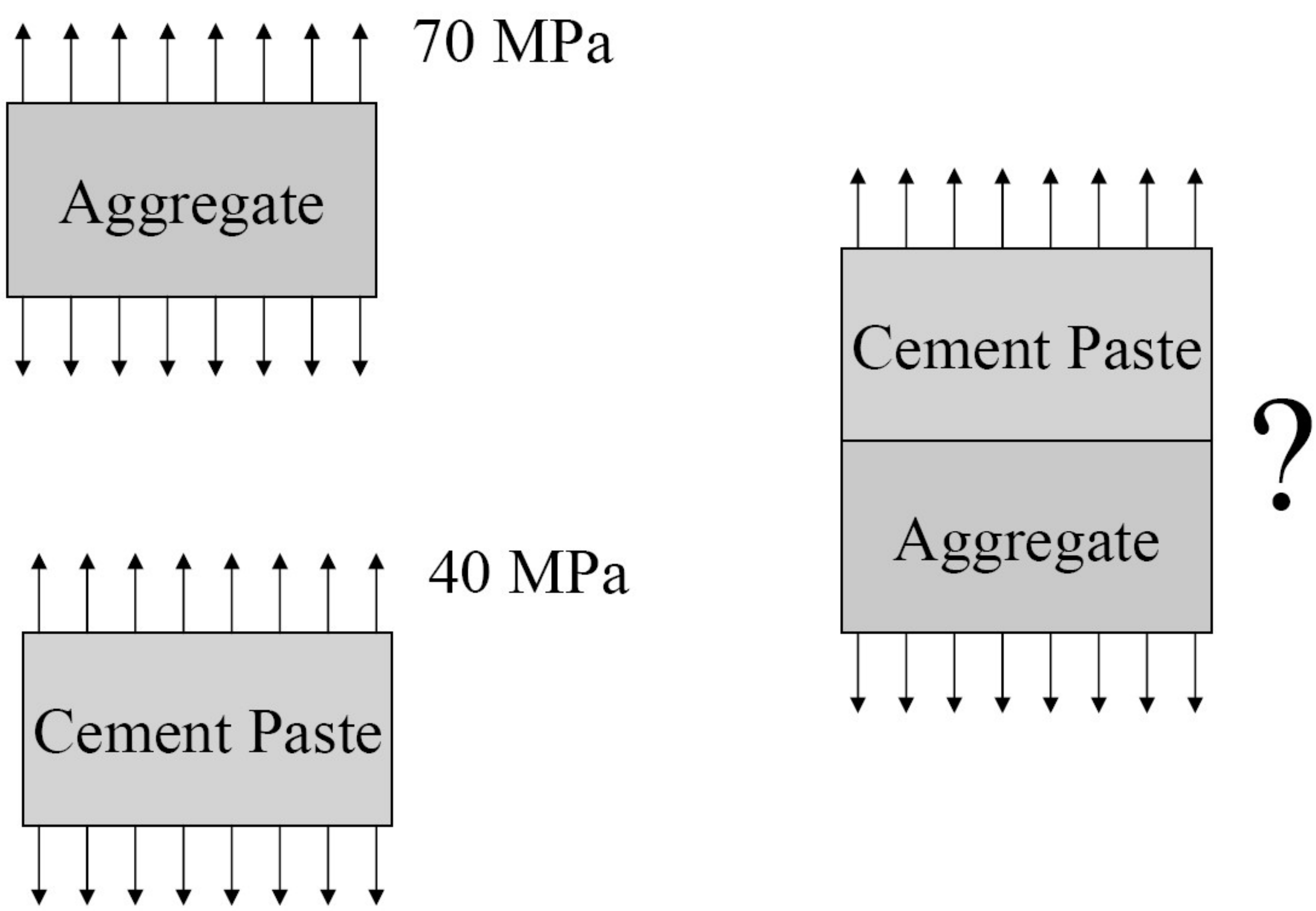


Aggregate Shape on Concrete Fresh Properties

- HIGHER SURFACE TO VOLUME RATIO
  - More cement paste required for coating aggregate surfaces
  - Less cement paste for lubrication and reduced workability of concrete
- IRREGULAR PARTICLES
  - Greater aggregate particles interaction
  - Reduced workability of concrete

Concrete is more workable with gravel aggregate compared to that with crushed aggregate, for a given mix proportion.

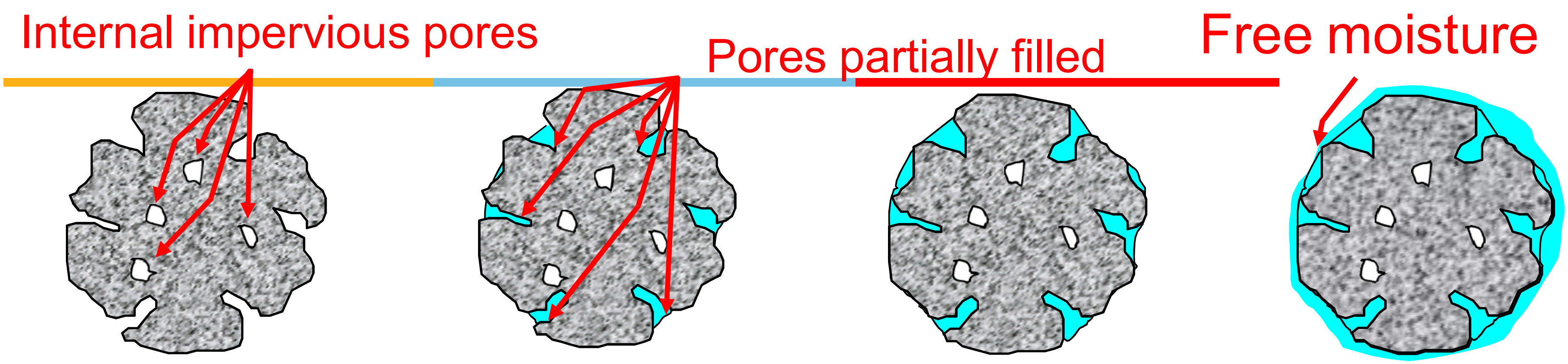
What Is the Strength of the Resulting Concrete?



Aggregate Strength

- Concrete strength is dominated by the weakest component either the paste, aggregate, or interface
  - Aggregate strength is of less importance for normal concrete
  - Aggregate strength is important in high-strength concrete and in the surface course on heavily traveled pavements
- Difficult and rare to test the strength of aggregate particles. Rather, tests on the **parent rock** (drilled cylinder, 1.5-2.5” in dia.) or a **bulk aggregate sample** as an indirect estimation of strength
- Typical compressive strength of 35 – 350 MPa

Absorption



Bone/oven dry – dried in oven to constant mass

W_{OD}

Air dry –

W_m

Moisture content

$$M = \frac{W_m - W_{OD}}{W_{OD}}$$

Saturated surface dry –

$W_{SSD} = W_{OD} + W_p$

Absorption

$$\mathbf{A} = \frac{W_{SSD} - W_{OD}}{W_{OD}}$$

Moist –

W_m

Moisture content

$$M = \frac{W_m - W_{OD}}{W_{OD}}$$

Absorption is the moisture content when the aggregates are in the SSD condition

Free moisture is the moisture content in excess of the SSD condition.

Percent free moisture = M - A

Important for proportioning concrete
negative free moisture – aggregates will absorb water
positive free moisture – aggregates will release water